

Module 2: Vector Calculus :

Scalar and Vector fields:

If every point (x, y, z) of a region R in space there corresponds a scalar $\phi(x, y, z)$ then ϕ is called a scalar point function and a scalar field ϕ is defined in R .

Eg:- i) $\phi = x^2 + y^2 + z^2$ ii) $\phi = xyz^3$.

If every point (x, y, z) of a region R in space there corresponds a vector $\vec{A}(x, y, z)$ then $\vec{A}$ is called a vector point function and a vector field $\vec{A}$ is defined in R .

Eg:- i) $\vec{A} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ ii) $\vec{A} = xyz\hat{i} + yz\hat{j} + z\hat{k}$.

* Gradient of a Scalar :

If ϕ is a scalar function, then the gradient of ϕ is a vector defined by

$$\text{grad } \phi \text{ or } \nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

Unit normal vector :

If $\phi(x, y, z) = c$ represents a surface then the unit normal vector to the surface is defined by

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

Directional derivative:

Directional derivative of ϕ along $\vec{a} = \nabla\phi \cdot \hat{a}$

* Maximum directional derivative = $|\nabla\phi|$

* Divergence of a vector:

If $\vec{f} = f_1\hat{i} + f_2\hat{j} + f_3\hat{k}$ then divergence of $\vec{f}$ is a scalar defined by

$$\text{div } \vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

Solenoidal vector: A vector $\vec{f}$ is said to be solenoidal iff $\text{div } \vec{f} = 0$.

* Curl of vector:

If $\vec{f} = f_1\hat{i} + f_2\hat{j} + f_3\hat{k}$ then the curl of a vector is a vector defined by

$$\text{curl } \vec{f} \text{ or } \nabla \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

$$\text{curl } \vec{f} = \hat{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) - \hat{j} \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} \right) + \hat{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right)$$

Irrotational Vector: A vector $\vec{f}$ is said to be irrotational iff $\text{curl } \vec{f} = 0$.

Problems:-

1) If $\phi = x^2 y^3 z^4$, find $\nabla\phi$, $|\nabla\phi|$ at $(1, -1, 1)$

Solⁿ:- Given $\phi = x^2 y^3 z^4$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

$$\nabla\phi = 2xy^3z^4 \hat{i} + 3x^2y^2z^4 \hat{j} + 4x^2y^3z^3 \hat{k}$$

At $(1, -1, 1)$

$$\nabla\phi = -2\hat{i} + 3\hat{j} - 4\hat{k}$$

$$|\nabla\phi| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$$

2) Find the unit normal vector to the surface $xy + yz + zx = c$ at the point $(-1, 2, 3)$.

Solⁿ:- $\nabla\phi = xy + yz + zx - c$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

$$\nabla\phi = (y+z)\hat{i} + (x+z)\hat{j} + (y+x)\hat{k}$$

At $(-1, 2, 3)$

$$\nabla\phi = 5\hat{i} + 2\hat{j} + \hat{k} \quad ; \quad |\nabla\phi| = \sqrt{25+4+1} = \sqrt{30}$$

$$\text{Unit normal vector : } \hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{5\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{30}}$$

3) Find the angle between the directions of the normal to the surface $x^2 yz = 1$ at the points $(-1, 1, 1)$ & $(1, -1, -1)$

Solⁿ:- Let $\phi = x^2 yz - 1$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

$$\nabla\phi = 2xyz\hat{i} + x^2\hat{j} + x^2y\hat{k}$$

i) At $(-1, 1, 1)$ $\nabla\phi = -2\hat{i} + \hat{j} + \hat{k}$.

$$|\nabla\phi| = \sqrt{6}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{-2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}} = \hat{n}_1$$

ii) At $(1, -1, -1)$ $\nabla\phi = 2\hat{i} - \hat{j} - \hat{k}$; $|\nabla\phi| = \sqrt{6}$

$$\hat{n}_2 = \frac{2\hat{i} - \hat{j} - \hat{k}}{\sqrt{6}}$$

$$\cos\theta = \hat{n}_1 \cdot \hat{n}_2 = \frac{(-2\hat{i} + \hat{j} + \hat{k})(2\hat{i} - \hat{j} - \hat{k})}{6} = \frac{-4 - 1 - 1}{6} = -1$$

$$\theta = 180^\circ = \pi$$

4) Find the angle between the surfaces $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at $(1, 1, 1)$

Soln:- Let $\phi = x \log z - y^2 + 1$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k} = \log z \hat{i} - 2y \hat{j} + x \hat{k}$$

At $(1, 1, 1)$ $\nabla\phi = -2\hat{j} + \hat{k}$; $|\nabla\phi| = \sqrt{5}$

$$\hat{n}_1 = \frac{-2\hat{j} + \hat{k}}{\sqrt{5}}$$

Let $\phi = x^2 y + z - 2$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k} = 2xy \hat{i} + x^2 \hat{j} + \hat{k}$$

At $(1, 1, 1)$ $\nabla\phi = 2\hat{i} + \hat{j} + \hat{k}$; $|\nabla\phi| = \sqrt{6}$

$$\hat{n}_2 = \frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}$$

$$\therefore \cos\theta = \hat{n}_1 \cdot \hat{n}_2 = \left(\frac{-2\hat{j} + \hat{k}}{\sqrt{5}} \cdot \frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}} \right)$$

$$\cos\theta = -\frac{1}{\sqrt{30}} \Rightarrow \theta = \cos^{-1}\left(-\frac{1}{\sqrt{30}}\right)$$

5) Find the directional derivative of $\phi = x^2 y z + 4x z^2$ at $(1, -2, -1)$ along $\vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Soln:- Let $\phi = x^2 y z + 4x z^2$

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

$$\nabla\phi = (2xy + 4z^2) \hat{i} + (xz^2) \hat{j} + (x^2 y + 8xz) \hat{k}$$

At $(1, -2, -1)$

$$\nabla\phi = 8\mathbf{i} - \mathbf{j} - 10\mathbf{k}.$$

Directional derivative (D.D) of ϕ along $\vec{a} = \nabla\phi \cdot \hat{a} = \nabla\phi \cdot \frac{\vec{a}}{|\vec{a}|}$

$$= (8\mathbf{i} - \mathbf{j} - 10\mathbf{k}) \cdot \left(\frac{2\mathbf{i} - \mathbf{j} - 2\mathbf{k}}{\sqrt{9}} \right)$$
$$= \frac{37}{3}.$$

6) In which direction the directional derivative of x^2yz^3 is maximum at $(2, 1, -1)$ and find the magnitude of this maximum.

Solⁿ:- Let $\phi = x^2yz^3$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = 2xyz^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$$

$$\nabla\phi_{(2,1,-1)} = -4\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}.$$

$$\text{Magnitude} = \sqrt{4^2 + 4^2 + 12^2} = 4\sqrt{11}. \quad (\text{Maximum D.D})$$

7) Find the maximum directional derivative of $\phi = xy + yz + zx$ at $(1, 1, 1)$.

Solⁿ: Given $\phi = xy + yz + zx$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k} = (y+z)\mathbf{i} + (x+z)\mathbf{j} + (x+y)\mathbf{k}.$$

$$\nabla\phi_{(1,1,1)} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}.$$

$$\text{Maximum D.D} = \sqrt{4+4+4} = \sqrt{12} = 2\sqrt{3}.$$

8) Find the divergence of $f = x^3z\mathbf{i} + y^3x\mathbf{j} + z^3y\mathbf{k}$.

Solⁿ:- $\vec{f} = x^3z\mathbf{i} + y^3x\mathbf{j} + z^3y\mathbf{k}$.

$$\text{div } \vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$\text{div } \vec{f} = 3x^2z + 3y^2x + 3z^2y.$$

9) Show that the vector $\vec{f} = 2xz\hat{i} - 10xyz\hat{j} + 3xz^2\hat{k}$ is solenoidal.

Solⁿ: $\vec{f} = 2xz\hat{i} - 10xyz\hat{j} + 3xz^2\hat{k}$.

$$\text{div } \vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} = 4xz - 10xz + 6xz = 0.$$

$\therefore \vec{f}$ is solenoidal.

10) Find $\text{curl } \vec{f}$ given that $\vec{f} = xyz^2\hat{i} + xy^2z\hat{j} + x^2yz\hat{k}$.

Solⁿ: Given, $\vec{f} = xyz^2\hat{i} + xy^2z\hat{j} + x^2yz\hat{k}$.

$$\text{curl } \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz^2 & xy^2z & x^2yz \end{vmatrix}$$

$$\text{curl } \vec{f} = \hat{i}(xz - xy^2) - \hat{j}(2xyz - 2xyz) + \hat{k}(y^2z - xz^2)$$

$$\text{curl } \vec{f} = \hat{i}(xz - xy^2) + \hat{k}(y^2z - xz^2).$$

11) Show that $\vec{f} = (4xy - z^3)\hat{i} + 2x^2\hat{j} - 3xz^2\hat{k}$ is irrotational.

Solⁿ: Given, $\vec{f} = (4xy - z^3)\hat{i} + 2x^2\hat{j} - 3xz^2\hat{k}$.

$$\text{curl } \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4xy - z^3 & 2x^2 & -3xz^2 \end{vmatrix}$$

$$\text{curl } \vec{f} = \hat{i}(0 - 0) - \hat{j}(-3z^2 + 3z^2) + \hat{k}(4x - 4x)$$

$$\text{curl } \vec{f} = 0.$$

$\therefore \vec{f}$ is irrotational.

12) If $\vec{A} = 2x^2\vec{i} - 3yz\vec{j} + xz^2\vec{k}$ and $\phi = 2x - x^3y$. Compute $\vec{A} \cdot \nabla\phi$ and $\vec{A} \times \nabla\phi$ at $(1, -1, 1)$.

Solⁿ:- $\phi = 2x - x^3y$.

$$\nabla\phi = \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k} ; -3x^2y\vec{i} - x^3\vec{j} + 2\vec{k}.$$

$$\nabla\phi(1, -1, 1) = 3\vec{i} - \vec{j} + 2\vec{k}.$$

$$\vec{A}(1, -1, 1) = 2\vec{i} + 3\vec{j} + \vec{k}.$$

$$\vec{A} \cdot \nabla\phi(1, -1, 1) = 3(2) + (-1)(3) + (2)(1) = 5$$

$$\vec{A} \times \nabla\phi = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ 3 & -1 & 2 \end{vmatrix} = \vec{i}(6+1) - \vec{j}(4-3) + \vec{k}(-2-9)$$

$$\vec{A} \times \nabla\phi = 7\vec{i} - \vec{j} - 11\vec{k}.$$

13) Find $\text{div}\vec{F}$ and $\text{curl}\vec{F}$ where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

Solⁿ:- $\phi = x^3 + y^3 + z^3 - 3xyz$

$$\vec{F} = \text{grad}\phi = \nabla\phi = \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}$$

$$\vec{F} = (3x^2 - 3yz)\vec{i} + (3y^2 - 3xz)\vec{j} + (3z^2 - 3xy)\vec{k}$$

$$\text{div}\vec{F} = \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy)$$

$$\text{div}\vec{F} = 6x + 6y + 6z = 6(x + y + z).$$

$$\text{curl}\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix}$$

$$\text{curl}\vec{F} = \vec{i}\{-3x - (-3x)\} - \vec{j}\{-3y - (-3y)\} + \vec{k}\{-3z - (-3z)\} =$$

$$\text{curl}\vec{F} = 0.$$

14) If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div}\vec{F}$ and $\text{curl}\vec{F}$ at the point $(1, -1, 1)$.

Soln:- Let $\phi = xy^3z^2$

$$\vec{F} = \nabla\phi = \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}$$

$$\vec{F} = y^3z^2\vec{i} + 3xy^2z^2\vec{j} + 2xy^3z\vec{k}$$

$$\text{div}\vec{F} = \nabla \cdot \vec{F}$$

$$= \frac{\partial}{\partial x}(y^3z^2) + \frac{\partial}{\partial y}(3xy^2z^2) + \frac{\partial}{\partial z}(2xy^3z)$$

$$\text{div}\vec{F} = 0 + 6xyz^2 + 2xy^3$$

$$\text{div}\vec{F}(1, -1, 1) = -8$$

$$\text{Curl}\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3z^2 & 3xy^2z^2 & 2xy^3z \end{vmatrix}$$

$$\text{Curl}\vec{F} = \vec{0}$$

15) If $\vec{A} = xz^3\vec{i} - 2x^2yz\vec{j} + 2yz^4\vec{k}$ find $\nabla \cdot \vec{A}$, $\nabla \times \vec{A}$ and $\nabla \cdot (\nabla \times \vec{A})$.

$$\text{Soln:- } \nabla \cdot \vec{A} = \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}\right)(xz^3\vec{i} - 2x^2yz\vec{j} + 2yz^4\vec{k})$$

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(xz^3) - \frac{\partial}{\partial y}(2x^2yz) + \frac{\partial}{\partial z}(2yz^4)$$

$$\nabla \cdot \vec{A} = z^3 - 2x^2z + 8yz^3$$

$$\nabla \cdot \vec{A}(1, -1, 1) = -9$$

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= i(2z^4 + 2x^2y) - j(0 - 3xz^2) + k(-4xyz - 0)$$

$$= 2(x^2y + z^4)i + 3xz^2j - 4xyzk$$

$$\nabla \times \vec{A}_{(1,-1,1)} = 3j + 4k$$

$$\nabla \cdot (\nabla \times \vec{A}) = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) (2(x^2y + z^4)i + 3xz^2j - 4xyzk)$$

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

16) Show that $\vec{f} = (y+z)i + (z+x)j + (x+y)k$ is irrotational. Also find a scalar function ϕ such that $\vec{f} = \nabla\phi$.

Solⁿ: $\vec{f} = (y+z)i + (z+x)j + (x+y)k$

$$\text{Curl } \vec{f} = \nabla \times \vec{f} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y+z & z+x & x+y \end{vmatrix}$$

$$\text{Curl } \vec{f} = i(1-1) - j(1-1) + k(1-1) = 0$$

$\therefore \vec{f}$ is irrotational.

To find ϕ such that $\vec{f} = \nabla\phi$

$$(y+z)i + (z+x)j + (x+y)k = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$$

Equating the corresponding components & integrating

$$\frac{\partial\phi}{\partial x} = y+z \quad \therefore \phi = \int (y+z) dx + \phi_1(y, z)$$

$$\phi = xy + xz + \phi_1(y, z)$$

$$\frac{\partial\phi}{\partial y} = z+x \quad \therefore \phi = \int (z+x) dy + \phi_2(x, z)$$

$$\phi = yz + xy + \phi_2(x, z)$$

$$\frac{\partial \phi}{\partial z} = x+y \quad \therefore \phi = \int (x+y) dz + \phi_3(x,y)$$

$$\phi = xz + yz + \phi_3(x,y)$$

Let us choose $\phi_3(x,y) = xy$; $\phi_2(x,z) = xz$, $\phi_1(y,z) = yz$

$$\therefore \underline{\phi = xy + yz + xz.}$$

17) Show that $\vec{F} = (z + \sin y)\vec{i} + (x \cos y - z)\vec{j} + (x - y)\vec{k}$ is irrotational. Hence find a scalar function ϕ such that $\vec{F} = \nabla \phi$.

Solⁿ:- Given $\vec{F} = (z + \sin y)\vec{i} + (x \cos y - z)\vec{j} + (x - y)\vec{k}$

$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z + \sin y & x \cos y - z & x - y \end{vmatrix}$$

$$\text{curl } \vec{F} = \vec{i}(-1+1) - \vec{j}(1-1) + \vec{k}(\cos y - \cos y)$$

$$\text{curl } \vec{F} = 0.$$

$\therefore \underline{\vec{F} \text{ is irrotational.}}$

To find ϕ such that $\vec{F} = \nabla \phi$.

$$(z + \sin y)\vec{i} + (x \cos y - z)\vec{j} + (x - y)\vec{k} = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

Equating the corresponding components and integrating

$$\frac{\partial \phi}{\partial x} = z + \sin y \Rightarrow \phi = zx + x \sin y + \phi_1(y, z)$$

$$\frac{\partial \phi}{\partial y} = x \cos y - z \Rightarrow \phi = x \sin y - zy + \phi_2(x, z)$$

$$\frac{\partial \phi}{\partial z} = x - y \Rightarrow \phi = xz - yz + \phi_3(x, y)$$

Let us choose $\phi_1(y, z) = -yz$; $\phi_2(x, z) = zx$, $\phi_3(x, y) = x \sin y$

$$\therefore \underline{\phi = x \sin y - zy + xz.}$$

18) Find the value of the constant 'a' such that the vector field, $\vec{F} = (axy - z^3)\vec{i} + (a-2)x^2\vec{j} + (1-a)xz^2\vec{k}$ is irrotational and hence find a scalar function ϕ such that $\vec{F} = \nabla\phi$.

Solⁿ:- $\vec{F} = (axy - z^3)\vec{i} + (a-2)x^2\vec{j} + (1-a)xz^2\vec{k}$.

$$\text{Curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (axy - z^3) & (a-2)x^2 & (1-a)xz^2 \end{vmatrix} = 0$$

$$\vec{i}(0-0) - \vec{j}((1-a)z^2 + 3z^2) + \vec{k}((a-2)2x - ax) = 0$$

$$(a-4)z^2\vec{j} + (a-4)x\vec{k} = 0$$

$$\therefore \underline{a=4}.$$

To find ϕ such that $\vec{F} = \nabla\phi$.

$$\frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k} = (4xy - z^3)\vec{i} + 2x^2\vec{j} - 3xz^2\vec{k}$$

$$\frac{\partial\phi}{\partial x} = 4xy - z^3 \quad \therefore \phi = 2x^2y - xz^3 + \phi_1(y, z)$$

$$\frac{\partial\phi}{\partial y} = 2x^2 \quad \therefore \phi = 2x^2y + \phi_2(x, z)$$

$$\frac{\partial\phi}{\partial z} = -3xz^2 \quad \therefore \phi = -xz^3 + \phi_3(x, y)$$

Let us choose $\phi_1(y, z) = 0$; $\phi_2(x, z) = -xz^3$,

$$\phi_3(x, y) = 2x^2y$$

$$\therefore \underline{\phi = 2x^2y - xz^3}$$

19) Show that $\vec{f} = 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$ is irrotational and find ϕ such that $\vec{f} = \nabla\phi$.

Solⁿ: $\text{Curl } \vec{f} = 0$; $\vec{f}$ is irrotational.

$$\phi = x^2yz^3 \dots$$

20) Find the constants a and b such that

$\vec{f} = (axy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (bxz^2 - y)\vec{k}$ is irrotational. Also find a scalar function ϕ such that $\vec{f} = \nabla\phi$.

Solⁿ: $a = 6$, $b = 3$.

$$\phi = 3x^2y + xz^3 - yz.$$

Curvilinear Coordinates :

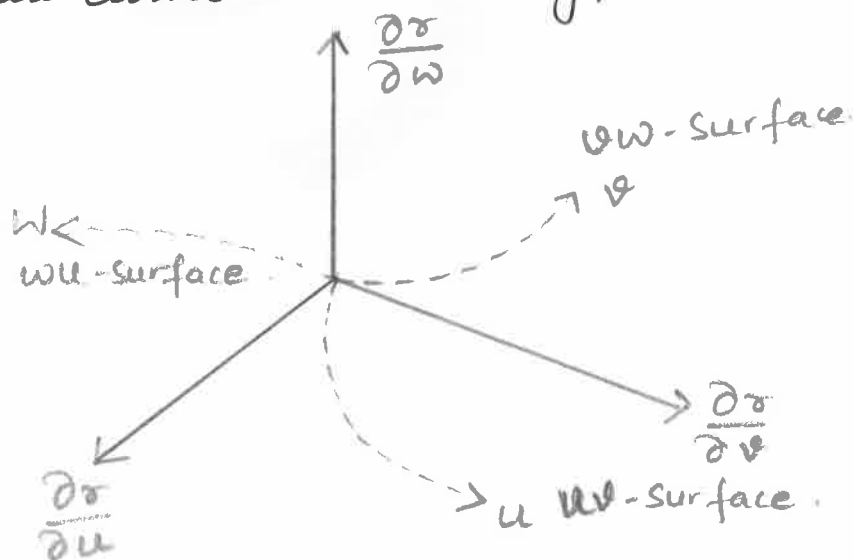
Let the coordinates of any point P in space be (x, y, z) in the Cartesian system.

Suppose x, y, z are expressible in terms of new coordinates (u_1, u_2, u_3) then x, y, z are functions of u_1, u_2, u_3 .

To express u_1, u_2, u_3 in terms of x, y, z by solving or eliminating then the coordinates (u_1, u_2, u_3) are known as Curvilinear Coordinates of the point P , where it is assumed that the correspondance between (x, y, z) and (u_1, u_2, u_3) is unique.

Orthogonal Curvilinear Coordinates :

A system of curvilinear coordinates is said to be orthogonal if at each point the tangents to the coordinate curves are mutually perpendicular.



* Scale factors, base vectors and orthogonality

Condition :

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of a point in space, we have $\vec{r} = \vec{r}(u_1, u_2, u_3)$.

$\frac{\partial \vec{r}}{\partial u_1}, \frac{\partial \vec{r}}{\partial u_2}, \frac{\partial \vec{r}}{\partial u_3} \rightarrow$ Tangent vectors.

$h_1 = \left| \frac{\partial \vec{r}}{\partial u_1} \right|, h_2 = \left| \frac{\partial \vec{r}}{\partial u_2} \right|, h_3 = \left| \frac{\partial \vec{r}}{\partial u_3} \right| \rightarrow$ Scale factors.

$$\hat{e}_1 = \frac{\frac{\partial \vec{r}}{\partial u_1}}{\left| \frac{\partial \vec{r}}{\partial u_1} \right|} = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial u_1}$$

$$\hat{e}_2 = \frac{\frac{\partial \vec{r}}{\partial u_2}}{\left| \frac{\partial \vec{r}}{\partial u_2} \right|} = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial u_2}$$

$$\hat{e}_3 = \frac{\frac{\partial \vec{r}}{\partial u_3}}{\left| \frac{\partial \vec{r}}{\partial u_3} \right|} = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial u_3}$$

Base vectors.

* Orthogonality of the curvilinear coordinate system

$$\rightarrow \hat{e}_1 \cdot \hat{e}_2 = 0, \hat{e}_2 \cdot \hat{e}_3 = 0, \hat{e}_3 \cdot \hat{e}_1 = 0$$

$$\rightarrow \hat{e}_1 \times \hat{e}_2 = \hat{e}_3, \hat{e}_2 \times \hat{e}_3 = \hat{e}_1, \hat{e}_3 \times \hat{e}_1 = \hat{e}_2$$

* Transformation between Cartesian and curvilinear systems:—

Let $u=x$, $v=y$, $w=z$ then $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$
 $\vec{r} = u\hat{i} + v\hat{j} + w\hat{k}$.

$$\frac{\partial \vec{r}}{\partial u} = \frac{\partial \vec{r}}{\partial x} = \hat{i} ; \quad \frac{\partial \vec{r}}{\partial v} = \frac{\partial \vec{r}}{\partial y} = \hat{j} ; \quad \frac{\partial \vec{r}}{\partial w} = \frac{\partial \vec{r}}{\partial z} = \hat{k}$$

The scale factors are

$$h_1 = \left| \frac{\partial \vec{r}}{\partial u} \right| = |\hat{i}| = 1, \quad h_2 = \left| \frac{\partial \vec{r}}{\partial v} \right| = |\hat{j}| = 1, \quad h_3 = \left| \frac{\partial \vec{r}}{\partial w} \right| = |\hat{k}| = 1$$

The base vectors are

$$e_1 = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial u} = \hat{i} ; \quad e_2 = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial v} = \hat{j} ; \quad e_3 = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial w} = \hat{k} .$$

For Cartesian co-ordinate system,

$$h_1 = h_2 = h_3 = 1 \quad \text{and} \quad e_1 = \hat{i}, e_2 = \hat{j}, e_3 = \hat{k} .$$

Also we have,

$$e_1 \cdot e_2 = 0 ; \quad e_2 \cdot e_3 = 0 ; \quad e_3 \cdot e_1 = 0 .$$

$$e_1 \times e_2 = e_3 ; \quad e_2 \times e_3 = e_1 ; \quad e_3 \times e_1 = e_2$$

Thus the Cartesian system is orthogonal.

* Scale factors of the cylindrical and spherical systems :-

Cylindrical System :-

Cylindrical polar coordinates (ρ, ϕ, z) given by the transformation: $x = \rho \cos \phi$; $y = \rho \sin \phi$; $z = z$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector.

$$\therefore \vec{r} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z\hat{k}.$$

By defⁿ of scale factors

$$h_1 = \left| \frac{\partial \vec{r}}{\partial \rho} \right| = |\cos \phi \hat{i} + \sin \phi \hat{j} + 0\hat{k}| = \sqrt{\cos^2 \phi + \sin^2 \phi} = 1.$$

$$h_2 = \left| \frac{\partial \vec{r}}{\partial \phi} \right| = |- \rho \sin \phi \hat{i} + \rho \cos \phi \hat{j} + 0\hat{k}| \\ = \sqrt{\rho^2 \sin^2 \phi + \rho^2 \cos^2 \phi} = \rho.$$

$$h_3 = \left| \frac{\partial \vec{r}}{\partial z} \right| = |0\hat{i} + 0\hat{j} + 1\hat{k}| = 1.$$

$\therefore h_1 = 1$; $h_2 = \rho$; $h_3 = 1$ are the scale factors for the cylindrical system.

Spherical System :-

Spherical polar coordinates (r, θ, ϕ) given by the transformation: $x = r \sin \theta \cos \phi$; $y = r \sin \theta \sin \phi$;

$$z = r \cos \theta.$$

Let $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ be the position vector.

$$\vec{r} = r \sin\theta \cos\phi \mathbf{i} + r \sin\theta \sin\phi \mathbf{j} + r \cos\theta \mathbf{k}.$$

By definition of scale factors,

$$h_1 = \left| \frac{\partial \vec{r}}{\partial r} \right| = \left| \sin\theta \cos\phi \mathbf{i} + \sin\theta \sin\phi \mathbf{j} + \cos\theta \mathbf{k} \right|$$

$$h_1 = \sqrt{\sin^2\theta (\cos^2\phi + \sin^2\phi) + \cos^2\theta}.$$

$$h_1 = \sqrt{\sin^2\theta + \cos^2\theta} = 1.$$

$$h_2 = \left| \frac{\partial \vec{r}}{\partial \theta} \right| = \left| r \cos\theta \cos\phi \mathbf{i} + r \cos\theta \sin\phi \mathbf{j} - r \sin\theta \mathbf{k} \right|$$

$$h_2 = \sqrt{r^2 \cos^2\theta (\cos^2\phi + \sin^2\phi) + r^2 \sin^2\theta}$$

$$h_2 = \sqrt{r^2 (\cos^2\theta + \sin^2\theta)} = r$$

$$h_3 = \left| \frac{\partial \vec{r}}{\partial \phi} \right| = \left| -r \sin\theta \sin\phi \mathbf{i} + r \sin\theta \cos\phi \mathbf{j} + 0 \mathbf{k} \right|$$

$$h_3 = \sqrt{r^2 \sin^2\theta (\sin^2\phi + \cos^2\phi)}$$

$$h_3 = r \sin\theta.$$

$\therefore$ $\boxed{h_1 = 1, h_2 = r, h_3 = r \sin\theta}$ for the spherical system.

* Find the scale factors for cylindrical system.

* Find the scale factors for spherical system.

* Orthogonality of the cylindrical system:

For the cylindrical system,

we have $\vec{r} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z \hat{k}$.

Let $\hat{e}_\rho, \hat{e}_\phi, \hat{e}_z$ be the basic vectors.

$$\therefore \hat{e}_\rho = \frac{\frac{\partial \vec{r}}{\partial \rho}}{\left| \frac{\partial \vec{r}}{\partial \rho} \right|} = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial \rho} = (\cos \phi \hat{i} + \sin \phi \hat{j} + 0 \hat{k}) \quad \therefore h_1 = 1$$

$$\hat{e}_\phi = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial \phi} = \frac{1}{\rho} (-\rho \sin \phi \hat{i} + \rho \cos \phi \hat{j} + 0 \hat{k}) \quad \therefore h_2 = \rho.$$

$$\hat{e}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j}$$

$$\hat{e}_z = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial z} = (0 \hat{i} + 0 \hat{j} + 1 \hat{k}) = \hat{k} \quad \therefore h_3 = 1$$

Consider,

$$\hat{e}_\rho \cdot \hat{e}_\phi = -\cos \phi \sin \phi + \sin \phi \cos \phi = \underline{0}.$$

$$\hat{e}_\phi \cdot \hat{e}_z = 0$$

$$\hat{e}_z \cdot \hat{e}_\rho = 0.$$

Thus the cylindrical system is orthogonal.

* Prove that the cylindrical coordinate system is orthogonal.

* Orthogonality of the spherical system :

For the spherical system, we have

$$\vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}.$$

Let $\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi$ be the basic vectors.

We have $h_1 = 1, h_2 = r, h_3 = r \sin \theta$.

Consider, $\hat{e}_r = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial r} = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}.$

$$\hat{e}_\theta = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial \theta} = \frac{1}{r} (r \cos \theta \cos \phi \hat{i} + r \cos \theta \sin \phi \hat{j} - r \sin \theta \hat{k})$$

$$\hat{e}_\theta = \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} - \sin \theta \hat{k}.$$

$$\hat{e}_\phi = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial \phi} = \frac{1}{r \sin \theta} (-r \sin \theta \sin \phi \hat{i} + r \sin \theta \cos \phi \hat{j} + 0 \hat{k})$$

$$\hat{e}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j}.$$

Consider,

$$\hat{e}_r \cdot \hat{e}_\theta = \sin \theta \cos \theta (\cos^2 \phi + \sin^2 \phi) - \sin \theta \cos \theta = 0.$$

$$\hat{e}_\theta \cdot \hat{e}_\phi = -\cos \theta \cos \phi \sin \phi + \cos \theta \sin \phi \cos \phi = 0$$

$$\hat{e}_\phi \cdot \hat{e}_r = -\sin \theta \cos \phi \sin \phi + \sin \theta \cos \phi \sin \phi = 0$$

Thus the spherical system is orthogonal.

* Prove that spherical polar coordinate system is orthogonal.

Problems:

1) Express the vector $\vec{A} = z\hat{i} - 2x\hat{j} + y\hat{k}$ in cylindrical coordinates.

Solⁿ:- Let $\vec{A} = A_1\hat{e}_\rho + A_2\hat{e}_\phi + A_3\hat{e}_z \rightarrow (1)$

Where A_1, A_2, A_3 are to be determined.

From (1), $\vec{A} \cdot \hat{e}_\rho = A_1$; $\vec{A} \cdot \hat{e}_\phi = A_2$; $\vec{A} \cdot \hat{e}_z = A_3 \rightarrow (2)$

In cylindrical system, $x = \rho \cos\phi$, $y = \rho \sin\phi$, $z = z$.

$\therefore \vec{A} = z\hat{i} - 2\rho \cos\phi\hat{j} + \rho \sin\phi\hat{k} \rightarrow (3)$

Also we have, $\hat{e}_\rho = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial \rho}$, $\hat{e}_\phi = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial \phi}$, $\hat{e}_z = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial z}$

Where $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$ and $\vec{r} = \rho \cos\phi\hat{i} + \rho \sin\phi\hat{j} + z\hat{k}$

Hence $\left. \begin{aligned} \hat{e}_\rho &= \cos\phi\hat{i} + \sin\phi\hat{j} \\ \hat{e}_\phi &= -\sin\phi\hat{i} + \cos\phi\hat{j} \\ \hat{e}_z &= \hat{k} \end{aligned} \right\} \rightarrow (4)$

Substituting (3) & (4) in (2) we get

$$A_1 = z \cos\phi - 2\rho \cos\phi \sin\phi = z \cos\phi - \rho \sin 2\phi$$

$$A_2 = -z \sin\phi - 2\rho \cos^2\phi = -(z \sin\phi + 2\rho \cos^2\phi)$$

$$A_3 = \rho \sin\phi.$$

$\therefore (1) \Rightarrow$

$$\boxed{\vec{A} = (z \cos\phi - \rho \sin 2\phi)\hat{e}_\rho - (z \sin\phi + 2\rho \cos^2\phi)\hat{e}_\phi + \rho \sin\phi \hat{e}_z.}$$

2) Express the vector $\vec{A} = 2x\hat{i} - 3y^2\hat{j} + xz\hat{k}$ in the cylindrical polar coordinate system.

Solⁿ:- Let $\vec{A} = A_1\hat{e}_r + A_2\hat{e}_\phi + A_3\hat{e}_z \rightarrow (1)$

Where A_1, A_2, A_3 are to be determined.

From (1), $\vec{A} \cdot \hat{e}_r = A_1$; $\vec{A} \cdot \hat{e}_\phi = A_2$; $\vec{A} \cdot \hat{e}_z = A_3 \rightarrow (2)$

In cylindrical system, $x = r\cos\phi$, $y = r\sin\phi$, $z = z$.

$\therefore \vec{A} = 2r\cos\phi\hat{i} - 3r^2\sin^2\phi\hat{j} + r\cos\phi z\hat{k} \rightarrow (3)$

Also we have $\hat{e}_r = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial r}$, $\hat{e}_\phi = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial \phi}$, $\hat{e}_z = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial z}$.

Where $h_1 = 1$, $h_2 = r$, $h_3 = 1$ and $\vec{r} = r\cos\phi\hat{i} + r\sin\phi\hat{j} + z\hat{k}$

Hence $\hat{e}_r = \cos\phi\hat{i} + \sin\phi\hat{j}$; $\hat{e}_\phi = -\sin\phi\hat{i} + \cos\phi\hat{j}$ $\rightarrow (4)$
 $\hat{e}_z = \hat{k}$.

Substituting (3) and (4) in (2) we get

$$A_1 = 2r\cos^2\phi - 3r^2\sin^3\phi$$

$$A_2 = -2r\sin\phi\cos\phi - 3r^2\sin^2\phi\cos\phi$$

$$A_3 = r\cos\phi z$$

$\therefore (1) \Rightarrow$

$$\vec{A} = (2r\cos^2\phi - 3r^2\sin^3\phi)\hat{e}_r - (2r\sin\phi\cos\phi + 3r^2\sin^2\phi\cos\phi)\hat{e}_\phi + (r\cos\phi z)\hat{e}_z.$$

3) Represent $\vec{F} = y\hat{i} - z\hat{j} + x\hat{k}$ in spherical polar coordinates and hence find F_r, F_θ, F_ϕ .

Solⁿ:- Let $\vec{F} = F_r \hat{e}_r + F_\theta \hat{e}_\theta + F_\phi \hat{e}_\phi \rightarrow (1)$

where F_1, F_2, F_3 are to be determined.

From (1), $\vec{F} \cdot \hat{e}_r = F_r, \vec{F} \cdot \hat{e}_\theta = F_\theta; \vec{F} \cdot \hat{e}_\phi = F_\phi \rightarrow (2)$.

In spherical system, $x = r \sin \theta \cos \phi$
 $y = r \sin \theta \sin \phi$
 $z = r \cos \theta$.

$\therefore \vec{F} = (r \sin \theta \sin \phi) \hat{i} - (r \cos \theta) \hat{j} + (r \sin \theta \cos \phi) \hat{k} \rightarrow (3)$

Also we have $\hat{e}_r = \frac{1}{h_1} \frac{\partial \vec{r}}{\partial r}; \hat{e}_\theta = \frac{1}{h_2} \frac{\partial \vec{r}}{\partial \theta}; \hat{e}_\phi = \frac{1}{h_3} \frac{\partial \vec{r}}{\partial \phi}$

Where $h_1 = 1, h_2 = r, h_3 = r \sin \theta$ and

$\vec{r} = (r \sin \theta \cos \phi) \hat{i} + (r \sin \theta \sin \phi) \hat{j} + (r \cos \theta) \hat{k}$

Hence $\hat{e}_r = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}$
 $\hat{e}_\theta = \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} - \sin \theta \hat{k}$
 $\hat{e}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j}$ } $\rightarrow (4)$

Substituting (3) and (4) in (2) we get

$F_r = r \sin^2 \theta \sin \phi \cos \phi - r \sin \theta \cos \theta \sin \phi + r \sin \theta \cos \theta \cos \phi$

$F_\theta = r \sin \theta \cos \theta \sin \phi \cos \phi - r \cos^2 \theta \sin \phi - r \sin^2 \theta \cos \phi$

$F_\phi = -r \sin \theta \sin^2 \phi - r \cos \theta \cos \phi$

$\therefore (1) =$

$\vec{F} = \{ r \sin \theta (\sin \theta \sin \phi \cos \phi - \cos \theta \sin \phi + \cos \theta \cos \phi) \} \hat{e}_r$
 $+ \{ r (\sin \theta \cos \theta \sin \phi \cos \phi - \cos^2 \theta \sin \phi - \sin^2 \theta \cos \phi) \} \hat{e}_\theta$
 $- \{ r (\sin \theta \sin^2 \phi + \cos \theta \cos \phi) \} \hat{e}_\phi.$